BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 109 · Binary Search

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Does linear search require a sorted list?

Q2. What is decomposition?
A. Splitting a problem into parts
B. Removing all useful details
C. Converting text to binary
D. Testing only the final program

Q3. “Every local variable can always be accessed directly from every other subprogram.” Correct this. Explain why.

Q4. Several procedures share a global score. “A global variable can be read only in one subprogram.” Correct this. What risk does sharing it create?

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I Do · Read new knowledge

Key vocabulary

Binary search — Halves a sorted search range after each unsuccessful comparison.

Sorted prerequisite — Values must be in a known order.

Midpoint — Middle index of current search interval.

Low/high — Inclusive bounds of the current search interval.

New learning

Sorting is essential: Binary search only works correctly when the array is ordered by the required comparison.

Start with a range: Keep low and high indexes of the active range, then calculate midpoint using integer division.

Discard half: If the target is larger than midpoint value, set low = mid+1; if smaller, set high = mid-1.

Stop correctly: Return the midpoint when equal; stop unsuccessful when low becomes greater than high.

Worked example

Trace binary search for 23 in [3,7,11,15,19,23,27,31]. Show low, mid and high. [5 marks]

1. Check precondition

2. The list is sorted; low=0 and high=7.

3. mid=(0+7) DIV 2=3; value 15 is less than 23, so low becomes 4.

4. mid=(4+7) DIV 2=5; value 23 matches. Found at index 5. [5]

We Do · Practise with guidance

Guided 1

Find 41 in [5,9,13,21,29,33,37,41,45] using binary search. [5]

Worked method: low=0, high=8, mid=4 gives 29 [1]; low=5, mid=6 gives 37 [2]; low=7, mid=7 gives 41 [1]; found at index 7 [1].

Guided 2

Explain why binary search cannot be relied upon for [9,2,7,1,5]. [3]

Worked method: Data is unsorted [1]; discarding halves depends on ordering [1], so the target might be discarded incorrectly [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Guided 3

Search for 6 in [2,4,8,10]. Explain when it stops. [4]

Worked method: low0 high3 mid1=4 [1]; low2 mid2=8 [1]; high1 [1], so low2>high1 and target is absent [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the midpoint assignment for a sorted list of bit weights.

```
array weights = [1,2,4,8,16,32,64]
low = 0
high = 6
mid = ...
print(weights[mid])
```

write

Question 1

A sorted array is [1,5,9,13,17,21,25]. Trace the mid indexes when searching for 21 and give its final index. [4] Source: Original OCR J277-style practice question.

Question 2

State the prerequisite for binary search and explain its importance. [3] Source: Original OCR J277-style practice question.

Question 3

Give the condition that shows binary search failed to locate an absent value. [2] Source: Original OCR J277-style practice question.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

Binary search can safely discard half only when data is sorted.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. No; it examines items in sequence until a match is found or the list ends.

Q2. A — Splitting a problem into parts

Q3. A local variable can be used only within its own subprogram or local scope.

Q4. Global variables can be used across the program. This risks unwanted changes and makes faults harder to trace. Use arguments and return values to avoid hidden changes.

I Do / We Do pages · Guided practice

Q1. low=0, high=8, mid=4 gives 29 [1]; low=5, mid=6 gives 37 [2]; low=7, mid=7 gives 41 [1]; found at index 7 [1].

Q2. Data is unsorted [1]; discarding halves depends on ordering [1], so the target might be discarded incorrectly [1].

Q3. low0 high3 mid1=4 [1]; low2 mid2=8 [1]; high1 [1], so low2>high1 and target is absent [1].

You Do pages · Code First

```
array weights = [1,2,4,8,16,32,64]
low = 0
high = 6
mid = (low + high) DIV 2
print(weights[mid])
```

Expected result: 8

You Do pages · Independent questions

Q1. Start low0 high6 mid3 value13 [1]; low4 high6 mid5 value21 [1]; target found [1] at index5 [1].

Q2. Sorted/ordered array [1]; comparisons with midpoint determine safe half to discard [1], without sorting the target could be excluded incorrectly [1].

Q3. low > high [1], meaning no active elements remain [1].

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Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: